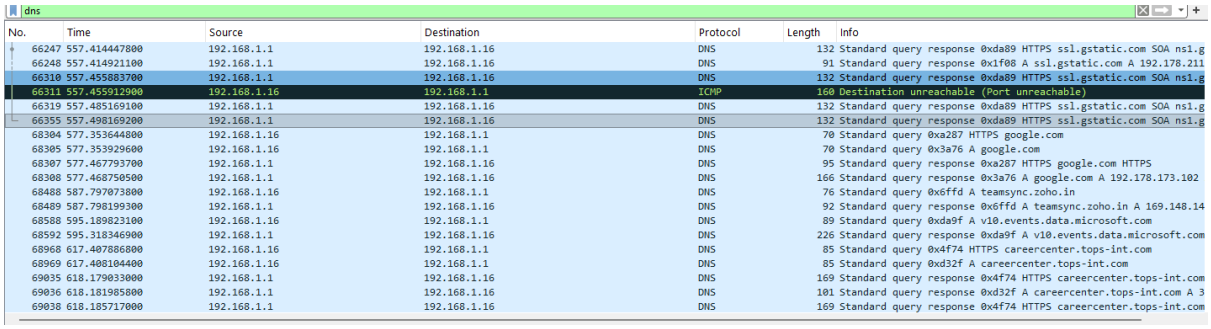
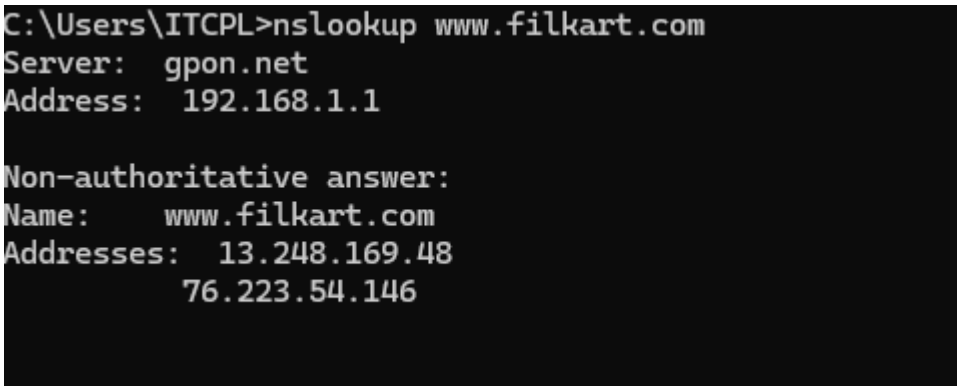
1) Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.



dns

No.	Time	Source	Destination	Protocol	Length	Info
66247	557.414447800	192.168.1.1	192.168.1.16	DNS	132	Standard query response 0xda89 HTTPS ssl.gstatic.com SOA ns1.g
66248	557.414921100	192.168.1.1	192.168.1.16	DNS	91	Standard query response 0x1f08 A ssl.gstatic.com A 192.178.211
66310	557.455883700	192.168.1.1	192.168.1.16	DNS	132	Standard query response 0xda89 HTTPS ssl.gstatic.com SOA ns1.g
66311	557.455912900	192.168.1.16	192.168.1.1	ICMP	160	Destination unreachable (Port unreachable)
66319	557.485169100	192.168.1.1	192.168.1.16	DNS	132	Standard query response 0xda89 HTTPS ssl.gstatic.com SOA ns1.g
66355	557.498169200	192.168.1.1	192.168.1.16	DNS	132	Standard query response 0xda89 HTTPS ssl.gstatic.com SOA ns1.g
68304	577.353644800	192.168.1.16	192.168.1.1	DNS	70	Standard query 0xa287 HTTPS google.com
68305	577.353929600	192.168.1.16	192.168.1.1	DNS	70	Standard query 0x3a76 A google.com
68307	577.467793700	192.168.1.1	192.168.1.16	DNS	95	Standard query response 0xa287 HTTPS google.com HTTPS
68308	577.468750500	192.168.1.1	192.168.1.16	DNS	166	Standard query response 0x3a76 A google.com A 192.178.173.102
68488	587.797073800	192.168.1.16	192.168.1.1	DNS	76	Standard query 0x6ffd A teamsync.zoho.in
68489	587.798199300	192.168.1.1	192.168.1.16	DNS	92	Standard query response 0x6ffd A teamsync.zoho.in A 169.148.14
68588	595.189823100	192.168.1.16	192.168.1.1	DNS	89	Standard query 0xda9f A v10.events.data.microsoft.com
68592	595.318346900	192.168.1.1	192.168.1.16	DNS	226	Standard query response 0xda9f A v10.events.data.microsoft.com
68968	617.407886800	192.168.1.16	192.168.1.1	DNS	85	Standard query 0x4f74 HTTPS careercenter.tops-int.com
68969	617.408104400	192.168.1.16	192.168.1.1	DNS	85	Standard query 0xd32f A careercenter.tops-int.com
69035	618.179033000	192.168.1.1	192.168.1.16	DNS	169	Standard query response 0x4f74 HTTPS careercenter.tops-int.com
69036	618.181985800	192.168.1.1	192.168.1.16	DNS	101	Standard query response 0xd32f A careercenter.tops-int.com A 3
69038	618.185717000	192.168.1.1	192.168.1.16	DNS	169	Standard query response 0x4f74 HTTPS careercenter.tops-int.com

2) Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.



```
C:\Users\ITCPL>nslookup www.filkart.com
Server:  gpon.net
Address:  192.168.1.1

Non-authoritative answer:
Name:    www.filkart.com
Addresses:  13.248.169.48
          76.223.54.146
```

IP address recorded: 76.223.54.146 (example)

Explanation: DNS (Domain Name System) converts a human-readable domain name, such as www.flipkart.com, into an IP address that computers use to locate the website's servers. Multiple IP addresses may be returned to support reliable and efficient content delivery.

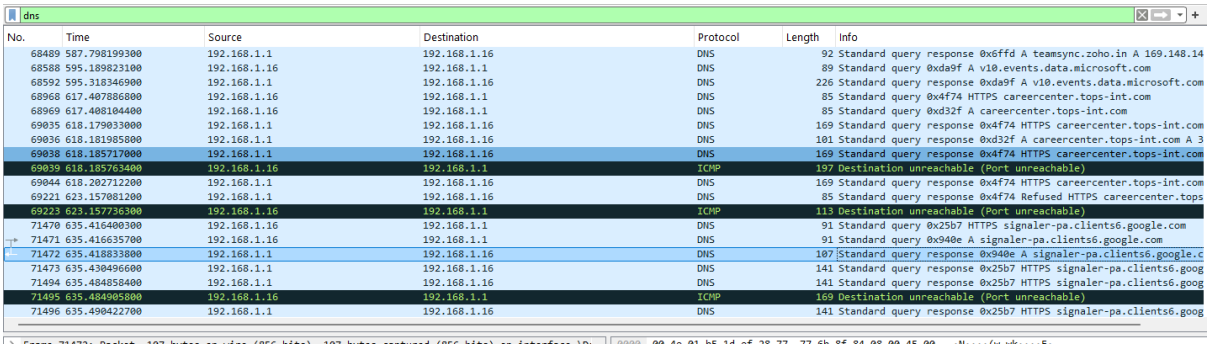
3) Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Difference Between HTTP and HTTPS

Feature	HTTP	HTTPS
1. Security	Data is sent without encryption.	Data is encrypted using TLS.
2. Data protection	Data can be intercepted and read more easily.	Protects data from being read or modified during transmission.
3. Website URL	Starts with http://	Starts with https:// and usually displays a padlock icon in the browser.

**4) Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.**



5) Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Protocol: DHCP (Dynamic Host Configuration Protocol)

Real-world scenario: Connecting to Wi-Fi to use Instagram

When I visit a café and connect my mobile phone to its Wi-Fi network to open Instagram, DHCP automatically assigns my phone an IP address. It also provides other network settings, such as the default gateway and DNS server.

This allows my phone to communicate with the internet without manually configuring network settings.

Conclusion: DHCP makes connecting to Wi-Fi easy by automatically assigning the necessary network settings to devices.